PHD125 Pharmaceutical Organic Chemistry · Lipids

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 30 minutes. Questions follow the subchapters of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. The answer key is at the back.

6.1–6.2 · Lipids and fatty acids

Q1 CH6-LO1

Give the shorthand for each fatty acid.

Options: C18:0 · C18:1 · C18:2 · C16:0 · C18:3

stearic acid _____

oleic acid _____

linoleic acid _____

palmitic acid _____

Q2 CH6-LO1

Which statement about natural unsaturated fatty acids is correct?

A. Their double bonds are usually cis, which kinks the chain

B. They are usually trans

C. They have odd numbers of carbons

D. They are branched

Q3 CH6-LO1

Name one of the two essential fatty acids.

Q4 CH6-LO1

EPA and DHA in fish oil are:

A. ω-3 fatty acids
B. ω-6 fatty acids
C. saturated fatty acids
D. steroids

6.3–6.4 · Waxes and triacylglycerols

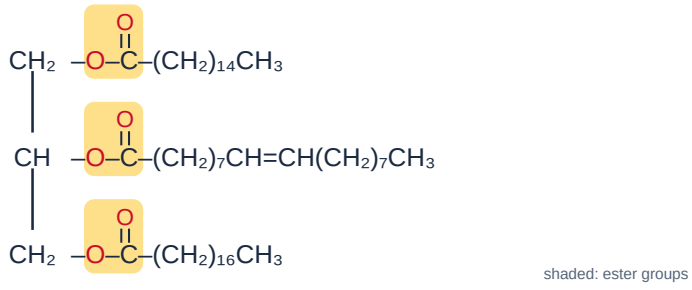
Q5 CH6-LO2

Carnauba wax is used in pharmacy to:

A. polish coated tablets
B. make soap
C. lower cholesterol
D. make suppositories melt

Q6 CH6-LO2

Identify this compound.



shaded: ester groups

A. a mixed triacylglycerol
B. a simple triacylglycerol
C. a wax
D. a phospholipid

Q7 CH6-LO3

Match each process to what happens.

Options: H_2 adds to C=C · O_2 breaks chains at C=C into aldehydes · water splits esters, releasing free fatty acids · fat turns into a steroid

hydrogenation _____

oxidative rancidity _____

hydrolytic rancidity _____

Q8 CH6-LO2

An oil with a higher iodine value has:

A. more C=C bonds
B. longer chains
C. more glycerol
D. fewer ester groups

6.5 · Hydrolysis

Q9 CH6-LO3

Pancreatic lipase hydrolyses a triglyceride to:

A. glycerol and fatty acids
B. glycerol and soap
C. cholesterol
D. glucose

Q10 CH6-LO3

How many moles of NaOH are needed to saponify one mole of a triacylglycerol?

_____

Q11 CH6-LO3

In a soap micelle, the nonpolar tails point outwards into the water.

True / False

6.6–6.7 · Phospholipids and membranes

Q12 CH6-LO4

Lecithin is:

A. phosphatidylcholine
B. a steroid
C. a wax
D. a triglyceride

Q13 CH6-LO4

Match each transport mode to an example.

Options: O_2 entering a cell · glucose through a GLUT protein · the Na^+/K^+ pump · a protein swallowed whole

simple diffusion _____

facilitated diffusion _____

active transport _____

Q14 CH6-LO4

Which drug form crosses lipid membranes most easily?

A. the uncharged free base
B. the hydrochloride ion
C. a large protein
D. a sugar

6.8 · Steroids

Q15 CH6-LO5

The steroid nucleus has:

A. three six-membered rings and one five-membered ring, fused

B. four benzene rings

C. a glycerol backbone

D. two fatty acids

Q16 CH6-LO5

Match each steroid to its role.

Options: raises blood glucose; anti-inflammatory · makes kidneys keep Na^+ · male sex hormone · bile acid for fat digestion

cortisol _____

aldosterone _____

testosterone _____

cholic acid _____

Q17 CH6-LO5

"Good" cholesterol refers to:

A. HDL, which carries cholesterol back to the liver

B. LDL

C. triglycerides

D. bile salts

Pharmacy scenario · Fats at the counter

Scenario

Puan Rosnah, 58, has just been prescribed **simvastatin**. She asks whether she should switch from palm oil to olive oil, whether fish-oil capsules will help, and whether her old hydrocortisone cream is "the same kind of steroid that bodybuilders use".

Q18 CH6-LO5

How does simvastatin lower her cholesterol?

A. It blocks HMG-CoA reductase, so her liver makes less cholesterol
B. It dissolves fat in the gut
C. It is a bile salt
D. It hydrogenates fats

Q19 CH6-LO2

Compared with palm oil, olive oil contains:

A. more unsaturated (oleic) and less saturated (palmitic) acid
B. more saturated fat
C. more trans fat
D. more cholesterol

Q20 CH6-LO5

Hydrocortisone and anabolic steroids are:

A. both steroids, but different classes with different actions
B. exactly the same drug
C. both phospholipids
D. both waxes

Q21 CH6-LO5

Summarise your advice to Puan Rosnah.

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

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About 30 minutes. Questions follow the subchapters of the lecture and end with a pharmacy scenario.

Answer key

6.1–6.2 · Lipids and fatty acids

Q1 CH6-LO1

Give the shorthand for each fatty acid.

stearic acid → C18:0; oleic acid → C18:1; linoleic acid → C18:2; palmitic acid → C16:0

Carbons : double bonds (FGOB 23.2).

Q2 CH6-LO1

Which statement about natural unsaturated fatty acids is correct?

A. Their double bonds are usually cis, which kinks the chain

B. They are usually trans

C. They have odd numbers of carbons

D. They are branched

FGOB 23.2.

Q3 CH6-LO1

Name one of the two essential fatty acids.

linoleic acid

The body cannot make linoleic or α-linolenic acid.

Q4 CH6-LO1

EPA and DHA in fish oil are:

A. ω-3 fatty acids

B. ω-6 fatty acids

C. saturated fatty acids

D. steroids

First C=C at the third carbon from the CH_3 end.

6.3–6.4 · Waxes and triacylglycerols

Q5 CH6-LO2

Carnauba wax is used in pharmacy to:

A. polish coated tablets

B. make soap

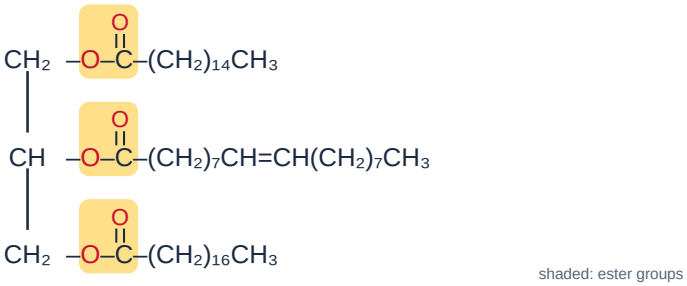
C. lower cholesterol

D. make suppositories melt

It gives tablets a shine.

Q6 CH6-LO2

Identify this compound.



shaded: ester groups

A. a mixed triacylglycerol

B. a simple triacylglycerol

C. a wax

D. a phospholipid

Glycerol with three different fatty acids.

Q7 CH6-LO3

Match each process to what happens.

hydrogenation → H_2 adds to C=C; oxidative rancidity → O_2 breaks chains at C=C into aldehydes; hydrolytic rancidity → water splits esters, releasing free fatty acids

Vitamin E and BHT slow oxidation.

Q8 CH6-LO2

An oil with a higher iodine value has:

A. more C=C bonds

B. longer chains

C. more glycerol

D. fewer ester groups

Iodine adds across C=C.

6.5 · Hydrolysis

Q9 CH6-LO3

Pancreatic lipase hydrolyses a triglyceride to:

A. glycerol and fatty acids

B. glycerol and soap

C. cholesterol

D. glucose

With water; base would give soap.

Q10 CH6-LO3

How many moles of NaOH are needed to saponify one mole of a triacylglycerol?

3

One per ester bond.

Q11 CH6-LO3

In a soap micelle, the nonpolar tails point outwards into the water.

True / False → **False**

Tails point inwards into the grease; heads face the water.

6.6–6.7 · Phospholipids and membranes

Q12 CH6-LO4

Lecithin is:

A. phosphatidylcholine

B. a steroid

C. a wax

D. a triglyceride

An emulsifier from soy or egg.

Q13 CH6-LO4

Match each transport mode to an example.

simple diffusion → O_2 entering a cell; facilitated diffusion → glucose through a GLUT protein; active transport → the Na^+/K^+ pump

Only active transport uses ATP.

Q14 CH6-LO4

Which drug form crosses lipid membranes most easily?

A. the uncharged free base

B. the hydrochloride ion

C. a large protein

D. a sugar

Ions do not dissolve in the hydrophobic core.

6.8 · Steroids

Q15 CH6-LO5

The steroid nucleus has:

A. three six-membered rings and one five-membered ring, fused

B. four benzene rings

C. a glycerol backbone

D. two fatty acids

Rings A, B, C, D.

Q16 CH6-LO5

Match each steroid to its role.

cortisol → raises blood glucose; anti-inflammatory; aldosterone → makes kidneys keep Na^+; testosterone → male sex hormone; cholic acid → bile acid for fat digestion

FGOB 23.6.

Q17 CH6-LO5

"Good" cholesterol refers to:

A. HDL, which carries cholesterol back to the liver

B. LDL

C. triglycerides

D. bile salts

High HDL lowers heart-disease risk.

Pharmacy scenario · Fats at the counter

Q18 CH6-LO5

How does simvastatin lower her cholesterol?

A. It blocks HMG-CoA reductase, so her liver makes less cholesterol

B. It dissolves fat in the gut

C. It is a bile salt

D. It hydrogenates fats

Statins lower LDL.

Q19 CH6-LO2

Compared with palm oil, olive oil contains:

A. more unsaturated (oleic) and less saturated (palmitic) acid

B. more saturated fat

C. more trans fat

D. more cholesterol

Olive oil is about 71% oleic acid; palm oil about 45% palmitic.

Q20 CH6-LO5

Hydrocortisone and anabolic steroids are:

A. both steroids, but different classes with different actions

B. exactly the same drug

C. both phospholipids

D. both waxes

Hydrocortisone is a corticosteroid; anabolic steroids are androgens.

Q21 CH6-LO5

Summarise your advice to Puan Rosnah.

Take simvastatin as prescribed (it blocks cholesterol synthesis in the liver). Replacing some saturated fat with unsaturated oils such as olive oil, limiting fried and processed foods with trans fats, and regular exercise help. Fish oil (ω-3) mainly lowers triglycerides; check with her doctor before adding it. Hydrocortisone is a corticosteroid for inflammation, not an anabolic (androgen) steroid; use it thinly for short periods.

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